CHAPTER 11

Output and Costs

ECON101 - Prince Sultan University

* Always think from a sellers perspective when dealing with costs

Lower cost → higher **Profit**

Profit = Total Revenue - Total cost

TC is divided into 2 types:

(1) Total variable costs ↑ changes with quantity

(2) Total Fixed costs ↑ NOT affected by anything
e.g paying yearly contract to landlords

cost of production:

(1) wages → paying workers

(2) capital investment in machinery

(3) raw materials purchase → change in price frequently

Key Formulas

- TC = TFC + TVC
- Deriving TFC → TC - TVC
- Deriving TVC → TC - TFC

- ATC = $\frac{TC}{Q}$
- AFC = $\frac{TFC}{Q}$
- AVC = $\frac{TVC}{Q}$

Marginal cost = $\frac{\Delta TC}{\Delta Q}$

Marginal Product = $\frac{\Delta TP}{\Delta L}$

Average Product = $\frac{TP}{L}$

OVERVIEW

Learning objectives

A firm's goal is to **maximize profit**. To reach it, the firm needs to know how much it can produce with the resources it controls and what each level of output will cost. This chapter answers both questions for a period in which the firm cannot change the size of its plant.

1. Distinguish between the short run and the long run.
2. Explain and illustrate a firm's short-run product curves.
3. Explain and derive a firm's short-run cost curves.

OVERVIEW

Running example: Qamar Bakery

Qamar Bakery makes cakes in one rented kitchen. Its ovens and equipment cannot be changed quickly, but it can decide each day how many bakers to employ. To keep the example simple, labor is the bakery's only variable input.

Item	Amount
Plant: the kitchen, its ovens and its equipment	Fixed
Cost of the plant (rent and equipment)	960 SAR per day
Wage of a baker	360 SAR per day
Output	Cakes per day

DECISION TIME FRAMES

The short run

Some of a firm's decisions are costly or impossible to undo, while others can be changed from one day to the next. Economists sort them into two time frames.

> The **short run** is a time frame in which the quantity of at least one resource used in production is fixed.

- For most firms, the fixed resource is capital, known as the firm's **plant**. For Qamar Bakery, it is the kitchen with its ovens and equipment.
- In general, a firm can still vary its other resources, such as labor, ingredients and energy; in our example, only labor.
- Short-run decisions are easy to reverse. The bakery can add a baker for the Eid season and stop employing the extra baker afterward.

DECISION TIME FRAMES

The long run and sunk cost

The long run

The **long run** is a time frame in which the quantities of all resources can be varied, including the size of the plant. → fixed in short run

- Qamar Bakery could move to a larger kitchen or install more ovens.
- Long-run decisions are hard to reverse, because a plant, once bought or built, cannot easily be changed.

Sunk cost

A **sunk cost** is a cost that the firm has already paid and cannot recover.

- Suppose the bakery paid 25,000 SAR for a decorating counter built to fit its kitchen, and the counter has no resale value. Those 25,000 SAR are a sunk cost.
- Nothing the bakery decides today can bring that money back, so a sunk cost is **irrelevant** to the firm's current decisions.

→ short run: at least one factor of production is fixed → less time to adjust capital investment compared to long-term
+ easily reversible with variable inputs

→ Long run: all inputs are entirely variable
+ decisions are not easily reversible

SHORT-RUN TECHNOLOGY CONSTRAINT

Total, marginal and average product

In the short run, Qamar Bakery can raise its output only by employing more bakers. Three measures link output to the quantity of labor:

1. **Total product** (*TP*) is the maximum quantity that a given number of workers can produce in a given period.
2. The **marginal product of labor** (*MP*) is the change in total product when one more worker is employed and all other inputs stay the same: $MP = \Delta TP / \Delta L$, where L is the number of workers.
3. The **average product of labor** (*AP*) is total product divided by the number of workers: $AP = TP / L$.

Qamar Bakery's product schedule

Point	Bakers (L)	Total product (cakes per day)	Average product (cakes per baker)	Marginal product (cakes per extra baker)
A	0	0		
				12
B	1	12	12	
				18
C	2	30	15	
				10
D	3	40	13.3	
				8
E	4	48	12	
				6
F	5	54	10.8	

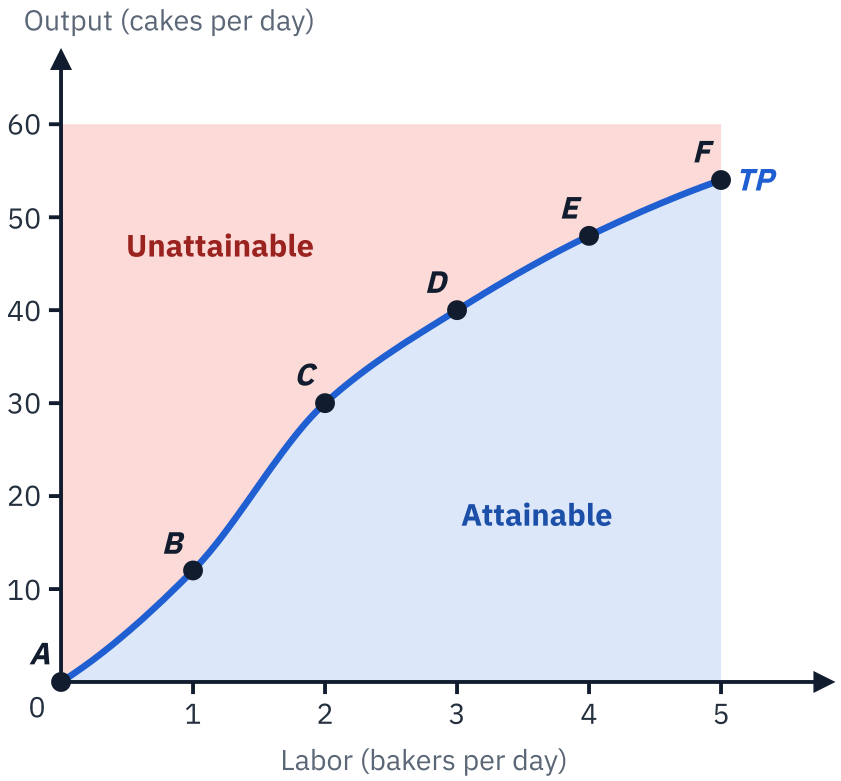
Each marginal product sits between two rows, because it is the change in total product from one quantity of labor to the next. For example, the 3rd baker raises output from 30 to 40 cakes, so the marginal product of going from 2 to 3 bakers is 10 cakes.

As the bakery employs more bakers:

- **Total product** keeps rising, from 12 to 54 cakes.
- **Marginal product** rises at first, from 12 to 18 cakes, and then falls.
- **Average product** rises at first, to its highest value of 15 cakes per baker with 2 bakers, and then falls.

SHORT-RUN TECHNOLOGY CONSTRAINT

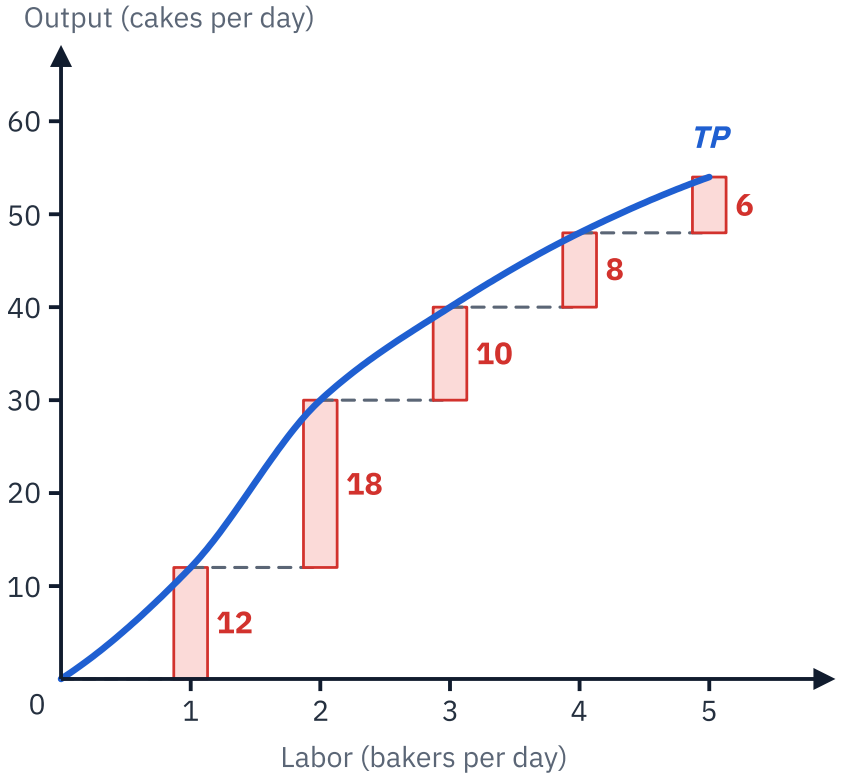
The total product curve



Product curves show how total, marginal and average product change as the firm employs more or less labor. The **total product curve** plots total product against the quantity of labor; points *A* to *F* come from the product schedule.

- The curve plays the same role as the production possibilities frontier: it **separates** the output levels the firm can reach in the short run from those it cannot.
- Points **on or below** the curve are **attainable**: 2 bakers can make up to 30 cakes a day.
- Points **above** it are **unattainable** with this plant: 2 bakers in this kitchen cannot make 40 cakes a day.
- The curve becomes steeper at first and flatter later. The next slides explain why.

Marginal product: steps in total product

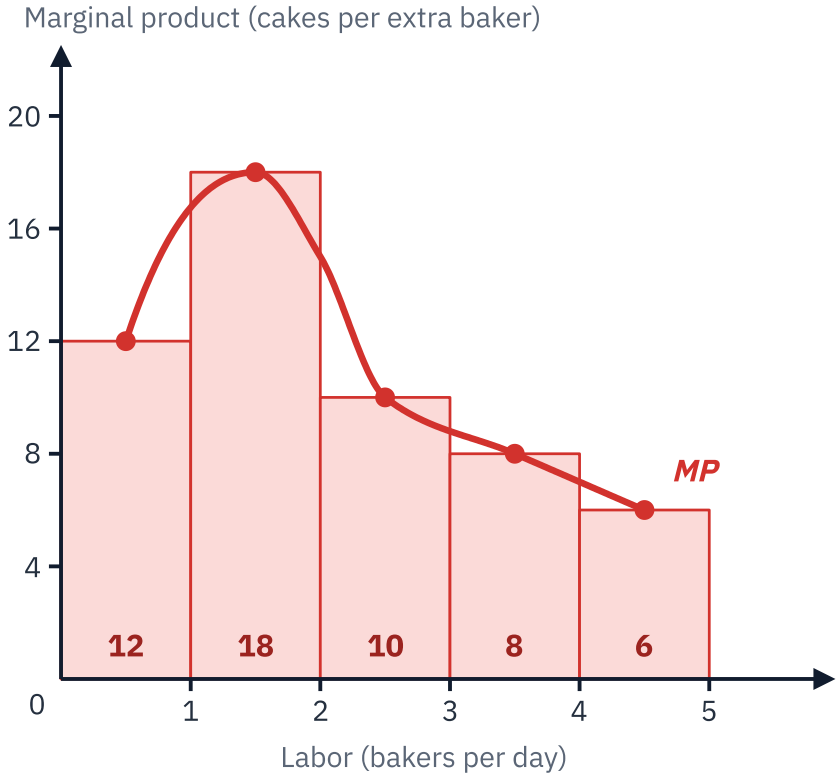


The marginal product of labor shows up in two ways: as the step in total product that each baker adds, and as a separate marginal product curve.

- Each red bar is the output added by one more baker. The 1st baker makes 12 cakes. The 2nd raises total product from 12 to 30 cakes, a marginal product of 18. The 3rd raises it from 30 to 40, a marginal product of 10, and so on.
- The tallest bar, 18 cakes, belongs to the 2nd baker. That is where the total product curve is steepest: marginal product is the slope of the *TP* curve.

↓ steeper slope → higher marginal product

The marginal product curve



- Standing the bars side by side gives each baker's marginal product.
- Each marginal product measures a change between two quantities of labor, such as from 1 to 2 bakers, so it is plotted halfway between them.
- The **marginal product curve** passes through these midpoints.

SHORT-RUN TECHNOLOGY CONSTRAINT

Increasing marginal returns

Almost every production process goes through two phases as the firm adds workers to a fixed plant.

① increasing MR ② diminishing MR

Increasing marginal returns occur when the marginal product of a worker is **greater** than the marginal product of the worker hired before.

- At Qamar Bakery, the 2nd baker adds 18 cakes, more than the 12 cakes added by the 1st.
- They come from **specialization and the division of labor**. One baker alone must mix, bake and decorate. With two, one can mix and bake while the other decorates, and neither loses time switching between tasks.

SHORT-RUN TECHNOLOGY CONSTRAINT

Diminishing marginal returns

Diminishing marginal returns occur when the marginal product of a worker is **smaller** than the marginal product of the worker hired before.

- At Qamar Bakery, the 3rd baker adds 10 cakes, fewer than the 18 cakes added by the 2nd. The 4th adds 8 and the 5th only 6.
- They arise because every extra baker shares the **same ovens and the same kitchen space**. With two bakers, the ovens are already busy most of the day. A 3rd baker can help mix and decorate but often waits for a free oven, so adds only 10 cakes. A 4th and a 5th find the ovens even busier and the counters more crowded, so each adds fewer cakes still.

↓ encounters more occupied resources → minimizes marginal productivity of each worker

SHORT-RUN TECHNOLOGY CONSTRAINT

The law of diminishing returns

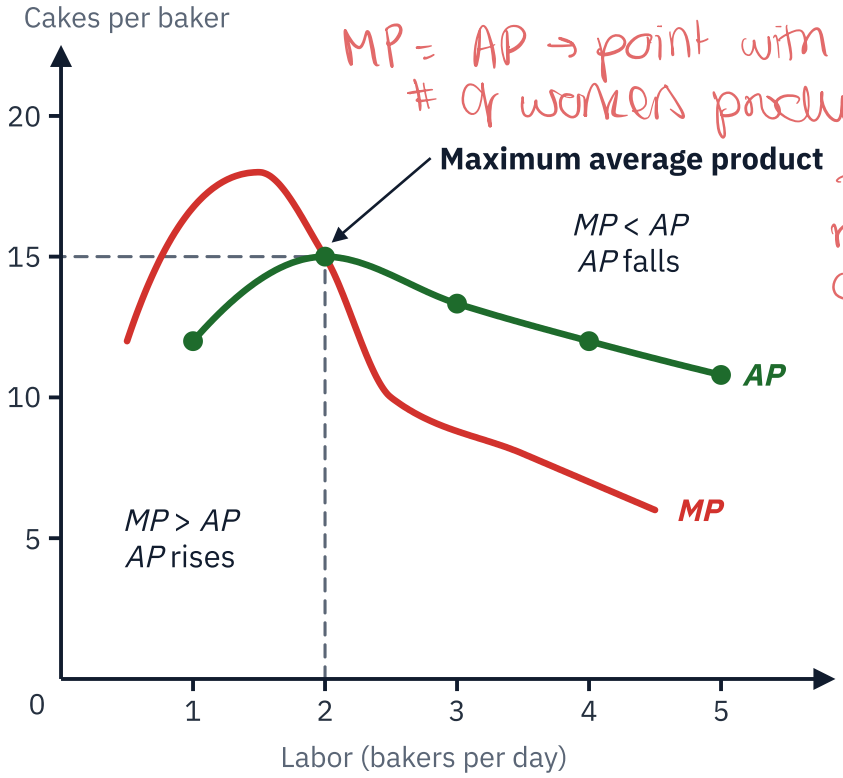
Diminishing marginal returns are found in so many production processes that economists call them a law.

The law of diminishing returns: as a firm uses more of a variable input with a given quantity of fixed inputs, the marginal product of the variable input eventually diminishes.

labor

At Qamar Bakery, the variable input is labor and the fixed input is the kitchen. Diminishing returns set in with the 3rd baker.

The average product curve



MP = AP → point with maximum # of workers producing the most output

The **average product curve** shows output per worker at each quantity of labor. It is drawn here together with the marginal product curve.

The marginal product of each new worker pulls the average toward itself.

The *MP* curve crosses the *AP* curve at the highest point of the *AP* curve, at 2 bakers.

How marginal product pulls the average

The marginal product of each new worker pulls the average toward itself:

MP above *AP*

When **marginal product exceeds average product**, average product **rises**. The 2nd baker adds 18 cakes, more than the 12-cake average of one baker, so the average rises to 15 cakes.

MP below *AP*

When **marginal product is below average product**, average product **falls**. The 3rd baker adds only 10 cakes, less than the average of 15, so the average falls to 13.3 cakes.

MP equals *AP*

When the **two are equal**, average product has reached its **maximum**. The *MP* curve crosses the *AP* curve at the highest point of the *AP* curve, at 2 bakers.

SHORT-RUN COST

Total cost

With its plant fixed, the bakery can bake more cakes only by paying for more bakers, so its costs climb as output grows. Economists use three measures of total cost.

Total cost (*TC*)

The cost of all the resources the firm uses.

Total fixed cost (*TFC*)

The cost of the firm's fixed inputs. It does **not** change with output. Qamar Bakery's total fixed cost is 960 SAR a day, even on a day when it bakes nothing.

Total variable cost (*TVC*)

The cost of the firm's variable inputs. It **does** change with output. Each baker costs 360 SAR a day, so $TVC = 360 \times L$.

Total cost is the sum of the other two: $TC = TFC + TVC$.

SHORT-RUN COST

Qamar Bakery's total costs

SAR per day

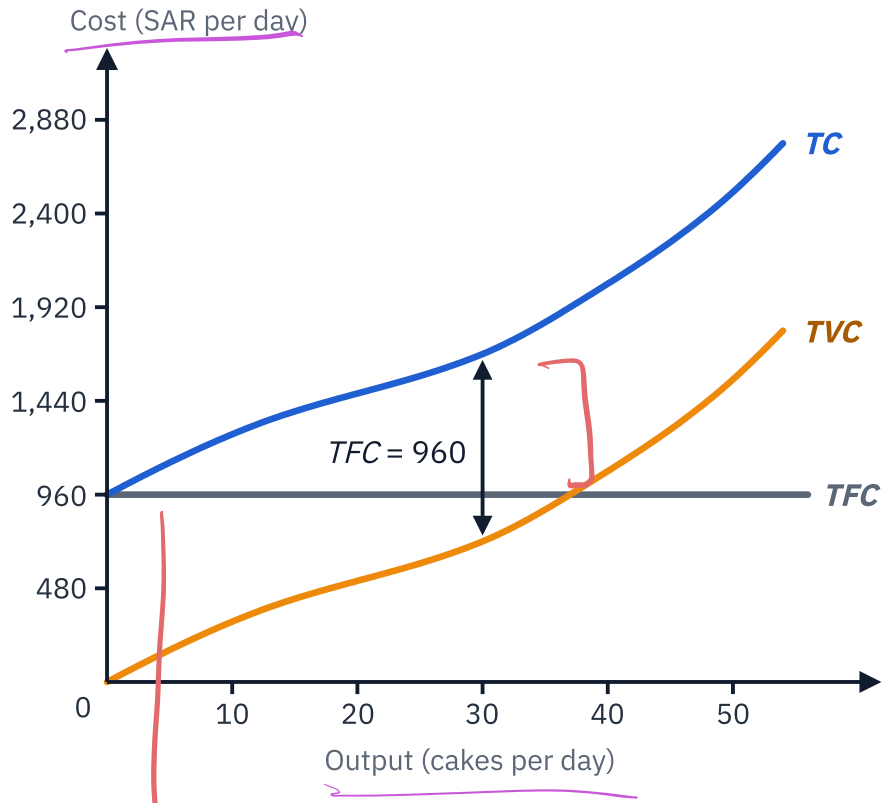
Bakers (*L*)	Output (cakes per day)	*TFC*	*TVC*	*TC*
0	0	960	0	960
1	12	960	360	1,320
2	30	960	720	1,680
3	40	960	1,080	2,040
4	48	960	1,440	2,400
5	54	960	1,800	2,760

960 SAR

- Total fixed cost is the same in every row.
- Total variable cost rises by 360 SAR with each baker, but the output that each baker adds is different. The 2nd baker adds 18 cakes, while the 5th adds only 6.
- Total cost rises by the same 360 SAR as total variable cost, because total fixed cost does not change.

Total cost curves

all cost curves and shapes are imp ★

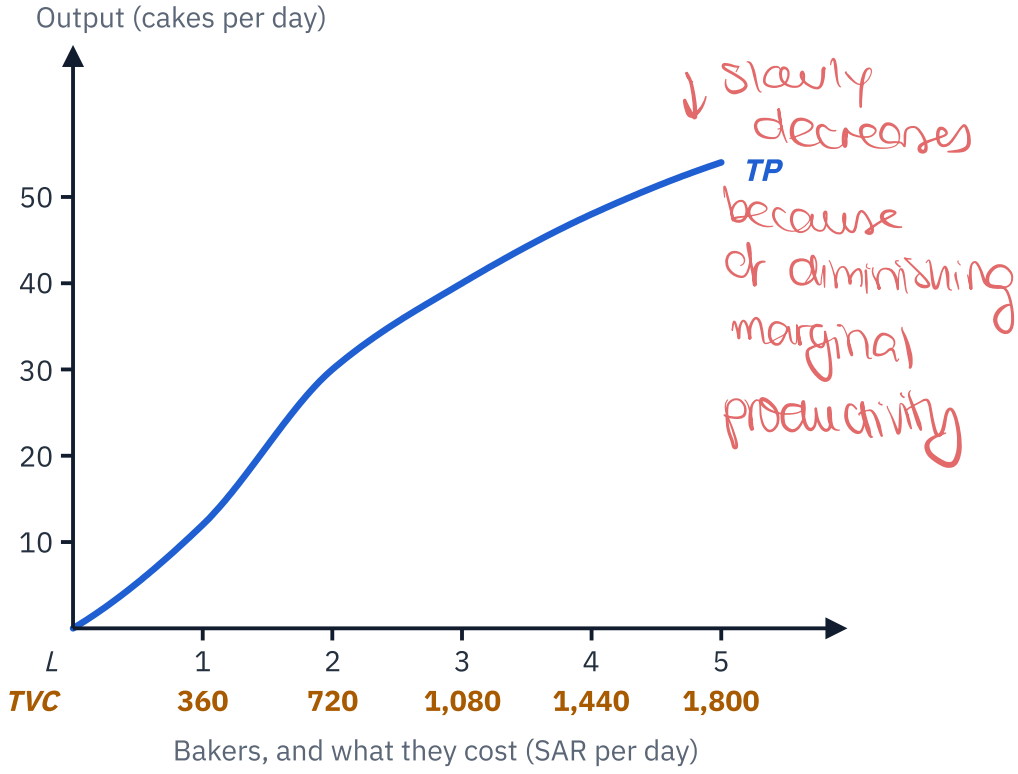


TFC is drawn as a horizontal line since cost never changes

The three total cost curves plot the costs in the table against output.

- The ***TFC* curve** is horizontal at 960 SAR, because fixed cost is the same at every output.
- The ***TVC* curve** slopes upward: more cakes require more bakers.
- The ***TC* curve** is the *TVC* curve shifted up by 960 SAR. The vertical gap between the two curves is always total fixed cost.
- The *TVC* and *TC* curves rise **slowly at first and steeply later**. From 12 to 30 cakes, each extra cake adds less to cost than before (20 SAR instead of 30), because the 2nd baker makes the team more productive. Beyond 30 cakes, diminishing returns make each extra cake more and more expensive.

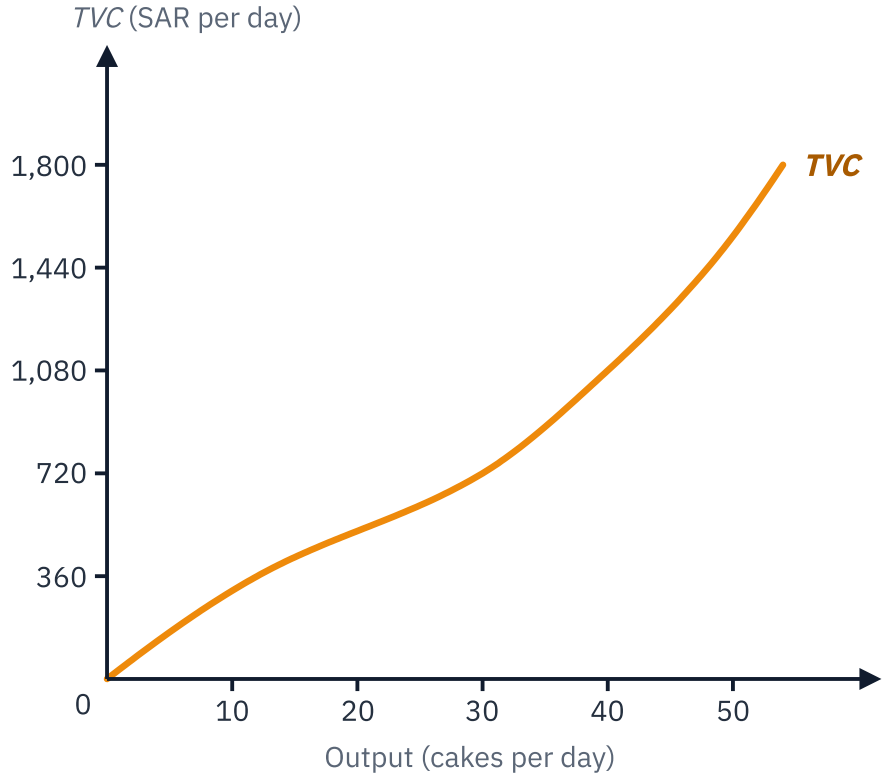
From the *TP* curve to the *TVC* curve



Each baker costs 360 SAR a day, so every quantity of labor matches one amount of total variable cost: 1 baker costs 360 SAR, 2 bakers cost 720 SAR, and so on.

- Under each number of bakers, write what those bakers cost.
- The total product curve now also shows how much output each amount of variable cost buys.

Swapping the axes: the *TVC* curve



- Swap the axes, so that output is on the horizontal axis and cost is on the vertical axis. The curve becomes the ***TVC* curve**. It is the total product curve **flipped over**.
- Where the *TP* curve gets **steeper** (increasing marginal returns), the *TVC* curve gets **flatter**. Where the *TP* curve gets **flatter** (diminishing marginal returns), the *TVC* curve gets **steeper**.
- Adding the 960 SAR of fixed cost to the *TVC* curve gives the *TC* curve shown earlier.

★ will be asked why it is cmp

SHORT-RUN COST

Marginal cost

Marginal cost (*MC*) is the increase in total cost that results from a one-unit increase in output:

$$MC = \Delta TC / \Delta Q$$

- The 2nd baker costs 360 SAR and adds 18 cakes, so each of those cakes adds 360/18 = 20 SAR to total cost.
- While marginal returns **increase**, each baker adds more cakes for the same wage, so marginal cost **falls**: from 30 to 20 SAR.
- Once marginal returns **diminish**, each baker adds fewer cakes, so marginal cost **rises**: to 36, 45 and then 60 SAR.

SHORT-RUN COST

Average cost

Each total cost has a matching average cost, which is that cost per unit of output (Q):

$$AFC = TFC/Q$$

Average fixed cost: total fixed cost per unit of output

$$AVC = TVC/Q$$

Average variable cost: total variable cost per unit of output

$$ATC = TC/Q$$

Average total cost: total cost per unit of output

$$ATC = AFC + AVC$$

Because $TC = TFC + TVC$, dividing by output gives $ATC = AFC + AVC$.

SHORT-RUN COST

Qamar Bakery's cost schedule (SAR)

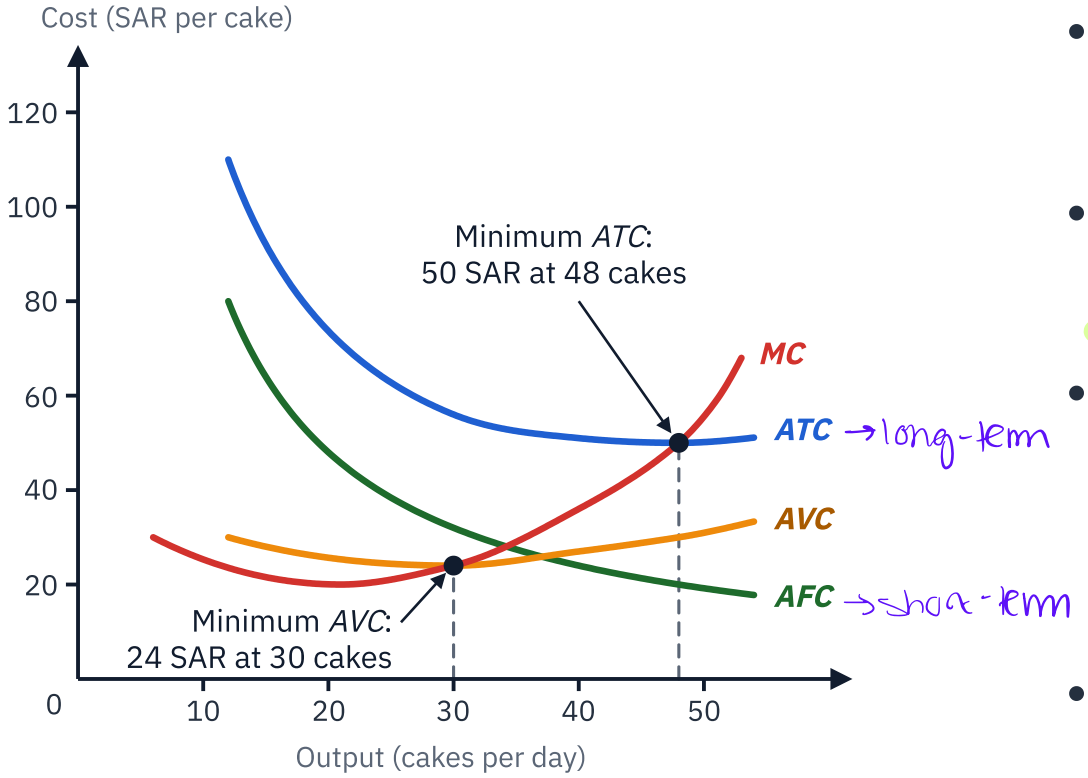
Values of AFC and AVC explains U-shape

Bakers	Output (*Q*)	*TC*	*AFC*	*AVC*	*ATC*	*MC* (per extra cake)
0	0	960				
						30
1	12	1,320	960/12 80	30	110	
						20
2	30	1,680	960/30 32	24	56	
						36
3	40	2,040	960/40 24	27	51	
						45
4	48	2,400	20	30	50	
						60
5	54	2,760	17.8	33.3	51.1	

Each marginal cost sits between two rows, because it is the cost of each extra cake when output rises from one row to the next. For example, from 40 to 48 cakes, (2,400 − 2,040)/(48 − 40) = 360/8 = 45 SAR. The last row is rounded to one decimal place.

Each *MC* value in the table is an average over the cakes that one baker adds. The *MC* curve on the next slides is a smooth illustration of how marginal cost changes cake by cake, so its lowest point, 20 SAR, sits at 21 cakes, midway through the 2nd baker's output.

Average and marginal cost curves



NEVER a straight line, it will continuously decrease

- ***AFC*** falls steadily as output rises, because the same 960 SAR is spread over more cakes.
- ***AVC*** is **U-shaped**. It falls to a minimum of 24 SAR at 30 cakes and then rises.
- ***ATC*** is also **U-shaped**, with a minimum of 50 SAR at 48 cakes. The vertical gap between the *ATC* and *AVC* curves is *AFC*, which shrinks as output grows.
- ***MC*** cuts the *AVC* curve and the *ATC* curve at their **minimum points**.

SHORT-RUN COST

MC cuts *AVC* and *ATC* at their minimums

MC below an average

Where *MC* is **below** an average cost, that average falls. The 2nd baker's cakes cost 20 SAR each, less than the *AVC* of 30 SAR at 12 cakes, so *AVC* falls to 24 SAR.

MC above an average

Where *MC* is **above** an average cost, that average rises. The 3rd baker's cakes cost 36 SAR each, more than 24 SAR, so *AVC* rises to 27 SAR.

MC equals an average

Where *MC* **equals** an average cost, that average is at its minimum: on the graph, *MC* meets *AVC* at 24 SAR (30 cakes) and *ATC* at 50 SAR (48 cakes). The same rule links *MC* and *ATC* in the table: the 4th baker's cakes cost 45 SAR, less than the *ATC* of 51 SAR, so *ATC* falls to 50 SAR; the 5th baker's cakes cost 60 SAR each, more than 50 SAR, so *ATC* rises again.

SHORT-RUN COST

Why the *AVC* curve is U-shaped

Average variable cost moves in the opposite direction to average product:

AP rises, *AVC* falls

At first, marginal product exceeds average product, so **average product rises** and ***AVC* falls**. With 2 bakers, each baker makes 15 cakes on average, so the wage cost per cake is 360/15 = 24 SAR.

AP falls, *AVC* rises

Later, marginal product falls below average product, so **average product falls** and ***AVC* rises**. With 4 bakers, each makes only 12 cakes on average, so *AVC* = 360/12 = 30 SAR.

Why the *ATC* curve is U-shaped

The *ATC* curve is the vertical sum of the *AFC* and *AVC* curves. Its U-shape comes from two forces pulling in opposite directions:

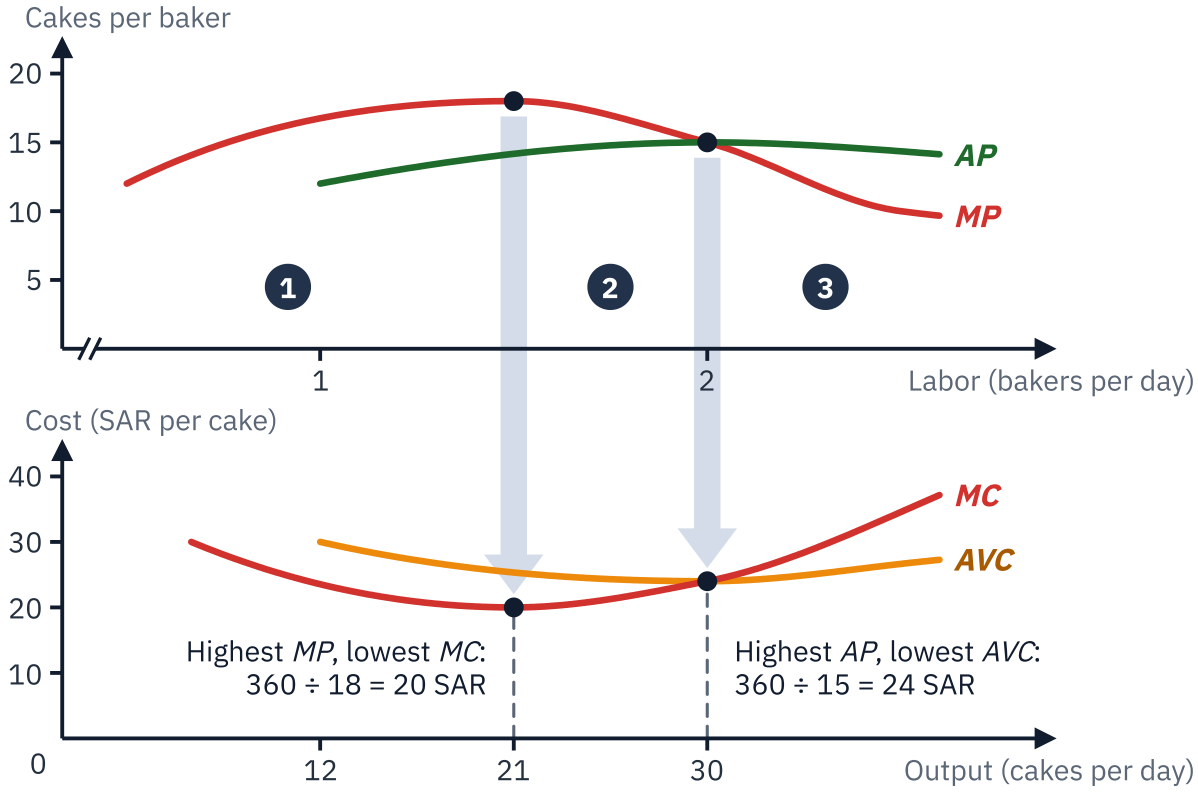
1. **Spreading total fixed cost over more output:** *AFC* falls as output rises.
2. **Eventually diminishing returns:** *AVC* rises, and at high output it rises faster than *AFC* falls.

Change in output	Change in *AFC*	Change in *AVC*	Change in *ATC*
30 to 40 cakes	−8	+3	−5
40 to 48 cakes	−4	+3	−1
48 to 54 cakes	−2.2	+3.3	+1.1

- Up to 48 cakes, the fall in *AFC* outweighs any rise in *AVC*, so *ATC* falls.
- At high output, *AFC* hardly falls, while diminishing returns keep pushing *AVC* up. Beyond 48 cakes the rise in *AVC* wins, and *ATC* rises.

share inverse relationship → mirror each other

Cost curves and product curves



Each number of bakers sits directly above the output it produces: 1 baker above 12 cakes, 2 bakers above 30 cakes.

1 *MP* rises, so *MC* falls. *AP* rises, so *AVC* falls.

2 *MP* falls, so *MC* rises. *AP* still rises, so *AVC* still falls.

3 *MP* and *AP* both fall, so *MC* and *AVC* both rise.

SHORT-RUN COST

Linking product and cost curves

A firm's technology, summarized by its product curves, determines the shape of its cost curves. The two sets of curves mirror each other.

Marginal product and marginal cost

Marginal cost is lowest where marginal product is highest. The 2nd baker has the largest marginal product, 18 cakes, and the cakes that baker adds have the lowest marginal cost, 20 SAR each. Over the range where *MP* rises, *MC* falls; where *MP* falls, *MC* rises.

Average product and average variable cost

AVC is lowest where average product is highest. With 2 bakers, average product peaks at 15 cakes per baker, and *AVC* reaches its minimum of 24 SAR at the matching output of 30 cakes. While *AP* rises, *AVC* falls. + vice versa

The link is the wage: *MC* = wage ÷ *MP* and *AVC* = wage ÷ *AP*. So the highest *MP* gives the lowest *MC*, 360 ÷ 18 = 20 SAR, and the highest *AP* gives the lowest *AVC*, 360 ÷ 15 = 24 SAR.

Shifts in cost curves: technology

Two things decide where a firm's cost curves sit: the firm's **technology** and the **prices of the factors of production** it buys.

- When new technology makes the bakers more productive, the **product curves move up** and the **cost curves move down**. If a faster mixer lets each baker make more cakes, every cake needs less labor and costs less.
- Some advances replace labor with capital. If the bakery leases an icing machine and employs fewer bakers, its **fixed cost rises** and its **variable cost falls**. *ATC* may then rise at low output, and it falls at high output.

Depends on FoP eg Cost
Saving Technology eg
Robotic Machines

Shifts in cost curves: factor prices

A higher price of a factor of production raises costs and shifts the cost curves. Which curves shift depends on whether the cost is fixed or variable.

- **A higher fixed cost** shifts the *TC* and *ATC* curves **upward** but leaves the *MC* curve **unchanged**. If a rent increase pushes total fixed cost from 960 to 1,200 SAR a day, *AFC* at 48 cakes rises from 20 to 25 SAR and *ATC* from 50 to 55 SAR. The cost of each extra cake does not change.
- **A higher variable cost** shifts the *TC*, *ATC* and *MC* curves **upward**, as well as *AVC*. If a baker's wage rises from 360 to 450 SAR a day, the 2nd baker's cakes cost 450/18 = 25 SAR each instead of 20 SAR, and *AVC* at 30 cakes rises from 24 to 450/15 = 30 SAR.

TVC (wage) / output per labor = AVC
↓15 since each worker at labor = 2 produces 15 on average
↓ refer back to AP = TP/L
productivity remains unchanged
→ across
2 bakes so 15 per baker

Change	*TC*	*ATC*	*AVC*	*MC*
Higher productivity	Down	Down	Down	Down
Higher fixed cost	Up	Up	No change	No change
Higher variable cost	Up	Up	Up	Up

SOURCE

Source

Michael Parkin (2022) *Microeconomics, Global Edition*, 14th Edition, published by Pearson Education.